MCA240102



PYTHON PROGRAMMING

Assignment - 2

1. Write a Python program to create a list of ten integers using the range() function. Perform the following operations:
   a) Display the list and access its elements using positive indexing, negative indexing, and slicing.
   b) Update an element at a specified index entered by the user.
   c) Perform list concatenation using + and repetition using *.
   d) Check whether a number exists in the list using in and not in.
   e) Perform the following list methods: append(), insert(), remove(), pop(), reverse(), and count().
   f) Display the list in ascending and descending order using sort() and sorted().
   g) Create an alias and a clone of the list, modify one element, and demonstrate the difference between aliasing and cloning.
   h) Display the maximum, minimum, and length of the list using max(), min(), and len().

2. Write a Python program to create a list containing the names of five students (list1)and a nested list containing the names and marks of at least five students(list2). Perform the following operations:
   a) Accept a student name from the user and search for it in the student list. Display its index if found; otherwise, display an appropriate message.
   b) Sort student names in ascending and descending alphabetical order using sort() and sorted().
   c) Add a new student name, insert a name at a specified position, and remove a student name.
   d) Display all student records using a loop(list2).
   e) Find the student who obtained the highest marks and calculate the average marks(list2).
   f) Display the number of student records using len()(list2).

3. Write a Python program to create a tuple containing at least ten values, including repeated numeric values. Perform the following operations:
   a) Display the tuple and access its elements using positive indexing, negative indexing, and slicing.
   b) Perform tuple concatenation using + and repetition using *.
   c) Check whether a specified element exists using in and not in.

d) Find the maximum, minimum, and length using max(), min(), and len().
e) Accept a value from the user and count its occurrences using count().
f) Search for a specified element using index(), displaying an appropriate message if it is not found.

4. Write a Python program to accept a string and another string from the user. Perform the following operations:

a) Display the strings and their lengths using len().
b) Access characters using positive and negative indexing.
c) Extract substrings using slicing.
d) Repeat a string using * and concatenate two strings using +.
e) Check whether a character or substring exists using in and not in.
f) Compare two strings using ==, !=, <, and >.
g) Convert strings into uppercase and lowercase using upper() and lower().

5. Write a Python program to accept a sentence from the user and perform the following operations:

Remove leading and trailing spaces using lstrip(), rstrip(), and strip().

a) Search for a substring using find() and index(). Handle cases where the substring is not found.
b) Count the occurrences of a specified word using count().
c) Replace a word or character using replace().
d) Split the sentence into words using split() and join them using join().
e) Demonstrate string immutability by attempting to modify a character through indexing, and explain why the operation is not permitted.
f) Sort the words alphabetically and display the sorted result.

6. Write a Python program using Python's array module to create an integer array containing at least eight elements. Perform the following operations:

a) Display the array and access elements using indexing and slicing.
b) Insert a new element at a specified position using insert().
c) Remove a specified element using remove() and remove the last element using pop().
d) Search for an element and display its index if found; otherwise, display an appropriate message.
e) Check element membership using in and not in.
f) Display the maximum, minimum, and length of the array.
g) Reverse the array and display the final result.